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Shenzhen Fuman Electronics Group Co., Ltd.

FM6124 (File No.: S&CIC1501)

SHEN ZHEN FINE MADE ELECTRONICS GROUP CO., LTD.

16-Channel Dual-Buffer Constant-Current Output LED Driver Chip

Overview

The FM6124 is a driver IC specifically designed for LED modules and displays, featuring 16-channel constant-current output capability. The FM6124 employs patented "output clamping" technology, which effectively eliminates the first-row dimness phenomenon while preventing LED damage.The FM6124 features an enhanced blanking function that delivers excellent blanking performance. It also offers superior immunity to interference, ensuring that its constant-current and low-gray-scale performance remain unaffected by the PCB. Additionally, the output stage current can be freely adjusted by selecting different external resistors, allowing for precise control of LED brightness.

During the display process (on the falling edge of OE), the FM6124 buffers 16-bit display data, allowing the system to continue writing 16-bit serial data while the FM6124 is displaying, resulting in a refresh rate that is more than 50% higher than that of general-purpose constant-current source chips.

The FM6124 incorporates precise current control technology, enabling on-chip error to be less than ±3.5% and inter-channel error to be less than ±2%.

Features

- 16 constant-current output channels
- Output current setting range:
  0.7–32 mA × 16 @ VDD = 5 V constant current output
  0.5–25 mA × 16@VDD=3.3V-channel constant current output
- Current Accuracy
  Current inconsistency between channels: ±1.25% (typical)
  ±2% (maximum)
  Current mismatch between chips: ±2% (typical)
  ±3.5% (maximum)
- Fast output current response (minimum): 30 ns @ VDD = 5 V
- I/O Schmitt Trigger Input
- Data transmission frequency: fMAX = 30 MHz (maximum)
- ESD HBM PASS 4 kV
- Supply Voltage: VDD = 3.3–6 V
- Operating temperature range: Topr = -40 to 85°C
- Features a function to reduce LED damage
- Features excellent blanking performance
- Effectively eliminates first-row dimness, low-gray-scale blocks, color cast in low-gray-scale areas, and speckling in low-gray-scale areas

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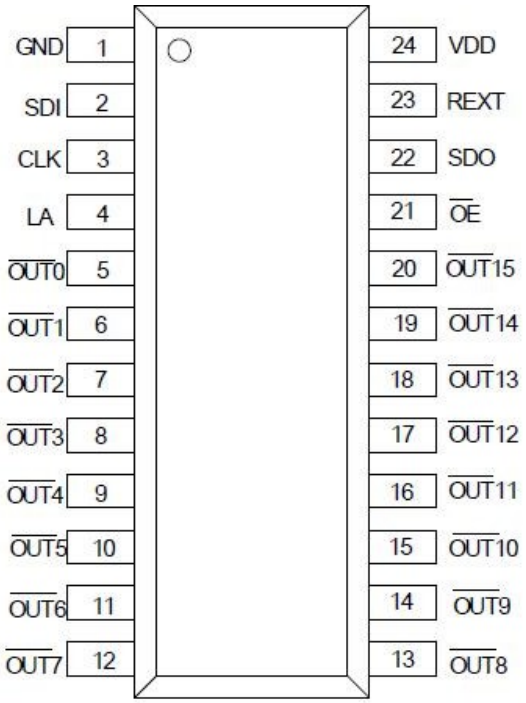
16-Channel Dual-Buffer Constant-Current Output LED Driver Chip

- Offers excellent anti-interference capability and low-grayscale performance
- Reduces the "caterpillar effect" caused by LED chip failure
- Integrated dual buffer, with a refresh rate over 50% higher than that of general-purpose constant-current chips
- Package Type: SSOP-24

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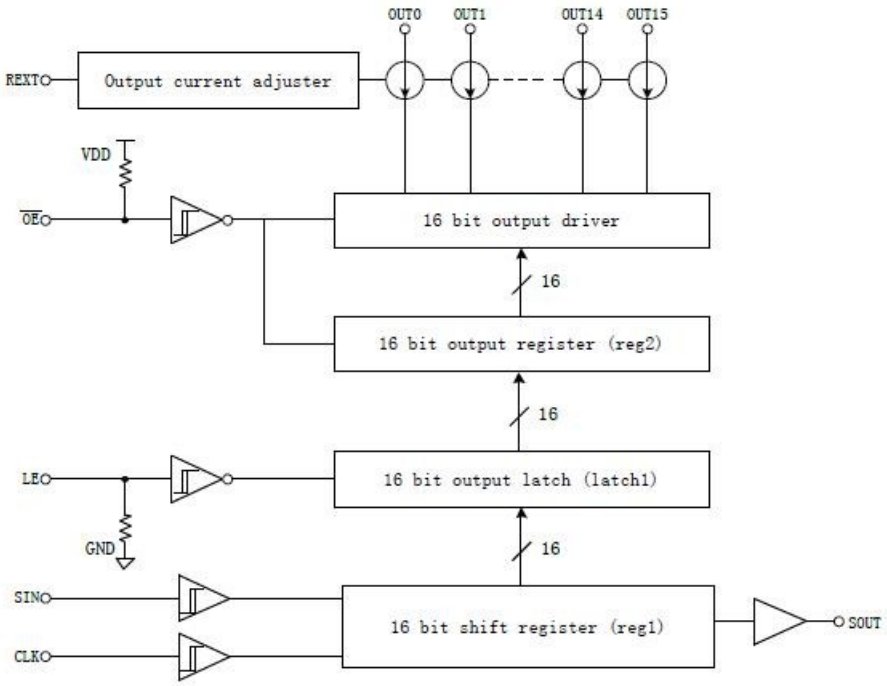
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Pin Definitions and Descriptions



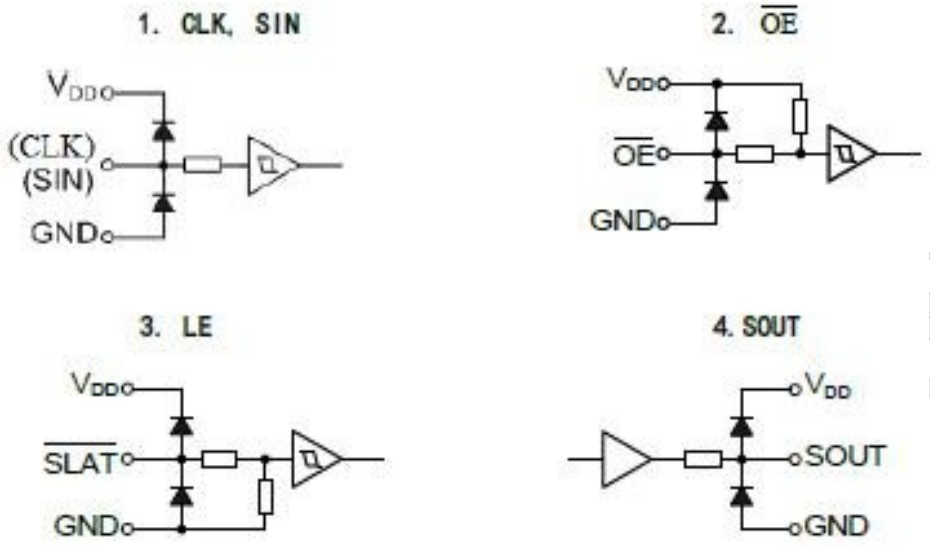
Pin Number	Pin Definition	Pin Name
1	GND	Chip Ground Pin
2	SDI	Serial data input to the shift register
3	CLK	Clock signal input pin
4	LA	Data latch input. When LE is high, data is transferred into the latch.
5–20	$\overline{OUT0}$—$\overline{OUT15}$	Constant Current Output Terminal
21	$\overline{OE}$	Output enable signal input; data is cached on the falling edge When OE is high, OUT0–OUT15 are disabled When OE is low, OUT0–OUT15 are enabled
22	SDO	Serial data output pin; can be connected to the SDI pin of the next driver chip
23	REXT	Output pin for an external adjustment resistor; used to adjust the output current
24	VDD	3.3V/5V Power Supply Input

Internal Block Diagram

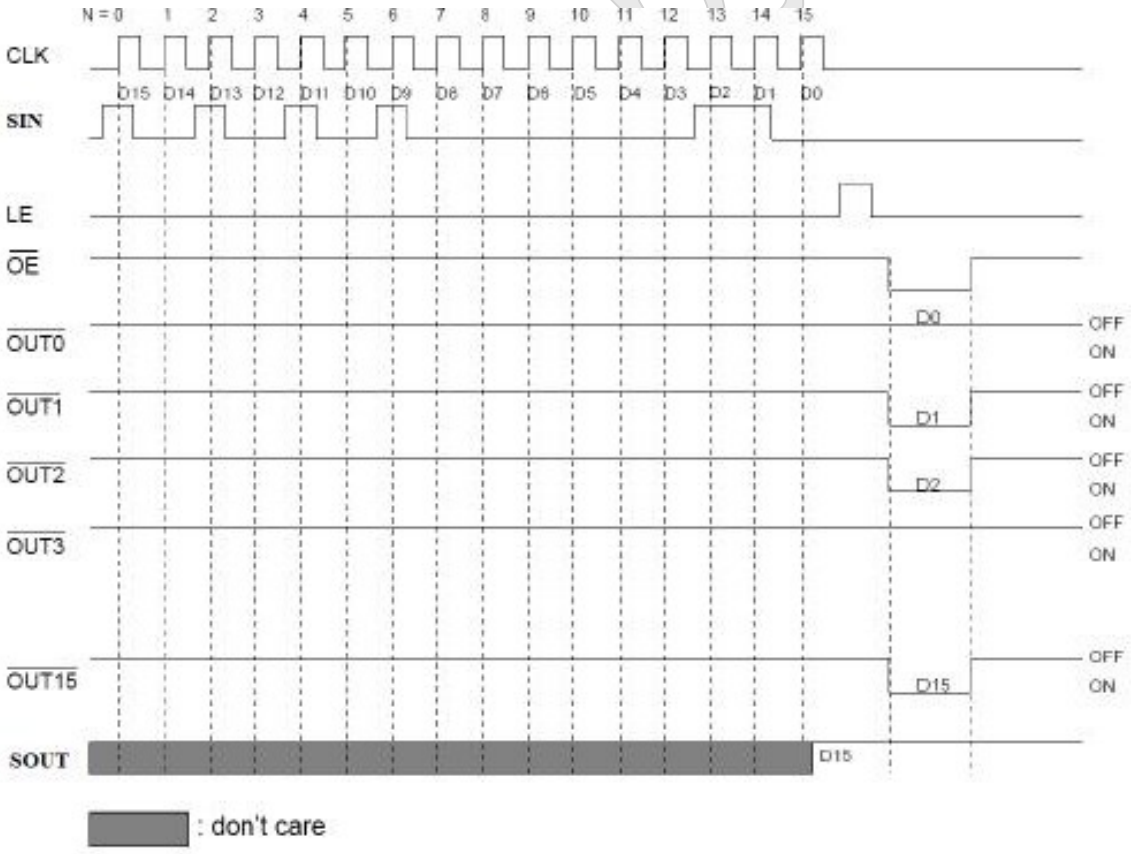


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I/O Equivalent Circuit



Timing Diagram



Truth Table

CLK	LE	$\overline{OE}$	SIN	$\overline{OUT0}$ … $\overline{OUT7}$ … $\overline{OUT15}$	SOUT
	H	L	Dn	DN`…DN-7…DN-15	DN-15
	L	L	Dn+1	No change	DN-14
	H	L	Dn+2	DN+2…DN-5…DN-13	DN-13
	X	L	Dn+3	DN+2…DN-5…DN-13	DN-13
	X	H	Dn+3	OFF	DN-13

Absolute Maximum Ratings (TA=25°C)

Characteristics	Symbol	Rated Value	Unit
Supply Voltage	VDD	0–7.0	V
Output Current	IO	32	mA
Input Voltage	VIN	-0.4 to VDD + 0.4	V
Output Withstand Voltage	VOUT	30	V
Clock Frequency	FCLK	30	MHz
Ground Current	IGND	512	mA
Power Consumption	PD	3	W
Thermal resistance	RTH (J-A)	39.15	°C/W
Operating Temperature	TOPR	-40–85	°C
Storage Temperature	TSTG	-55–150	°C

DC Characteristics (Unless otherwise specified, TA = 40°C–85°C)

Characteristics	Symbol	Test Conditions	Minimum Value	Typical Value	Maximum	Unit
Supply Voltage	VDD	-	3.3	5	6.0	V
Output voltage when ON	VO(ON)	$\overline{OUTn}$	0.6	-	4	V
High-level logic input voltage	VIH	-	0.7*VDD	-	VDD	V
Low-level logic input voltage	VIL	-	GND	-	0.3*VDD	V

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SOUT High-Level Output Current	IOH	VDD=5V	-	-1	-	mA
SOUT Low-Level Output Current	IOL	VDD=5V	-	1	-	mA
Constant-current output	IO	$\overline{OUTn}$	0.5	-	32	mA

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Dynamic Characteristics (Unless otherwise specified, VDD = 4.5–5.5 V, TA = 40°C–85°C)

Characteristics	Symbol	Test Circuit	Test Conditions	Minimum	Typical Value	Maximum	Unit
Serial Data Transmission Frequency	FCLK	6	-	-	-	30	MHz
Clock pulse width	TWCLK	6	SCK=H/L	20	-	-	nS
Buffer pulse width	TWLE	6	LE=H	20	-	-	nS
Enable pulse width	TWOE	6	$\overline{OE}$ =H/L, REXT=890Ω	30	-	-	nS
Hold time	THOLD1	6	-	5	-	-	nS
	THOLD2	6	-	5	-	-	nS
Creation Date	TSETUP1	6	-	5	-	-	nS
	TSETUP2	6	-	5	-	-	nS
Maximum clock rise time	TR	6		-	-	500	nS
Maximum clock fall time	TR	6		-	-	500	nS

Electrical Characteristics

Characteristics	Symbol	Test Circuit	Test Conditions	Minimum Value	Typical Value	Maximum	Unit
High-level logic Output Voltage	VOH	1	IOH = -1 mA, SOUT	VDD - 0.4	-	VDD	V
Low-level logic Output Voltage	VOL	1	IOH = +1 mA, SOUT	-	-	0.4	V
High-level logic Input current	IIH	2	VIN = VDD, OE, SIN, CLK	-	-	1	uA
Low-level logic Input current	IIL	3	VIN=GND, LE, SIN, CLK	-	-	-1	uA

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S&CIC1501)	IDD1	4	REXT = Not connected, OUT OFF	-	2.0	5.0	mA
Supply current	IDD2	4	REXT=1200，OUT OFF	-	5.5	9	mA
	IDD3	4	REXT=600，OUT OFF	-	6.5	10	mA
	IDD4	4	REXT=1200，OUT ON	-	8.2	12	mA
	IDD5	4	REXT=600，OUT ON	-	10	15	mA
Constant-current output	IO1	5	VDD=5.0V, VO=2.0V, REXT=1.19 kΩ	-	15	-	mA
	IO2	5	VDD=5.0V, VO=2.0V, REXT=595Ω	-	30	-	mA
Constant Current Error	△IO	5	VDD=5.0V, VO=2.0V, REXT=1.19 kΩ	-	±0.15	±0.37	mA
Constant-current power supply Voltage Regulation	%VDD	5	VDD=4.5–5.5 V VO = 2.0 V, REXT = 1.19 kΩ	-	±0.2	-	%/V

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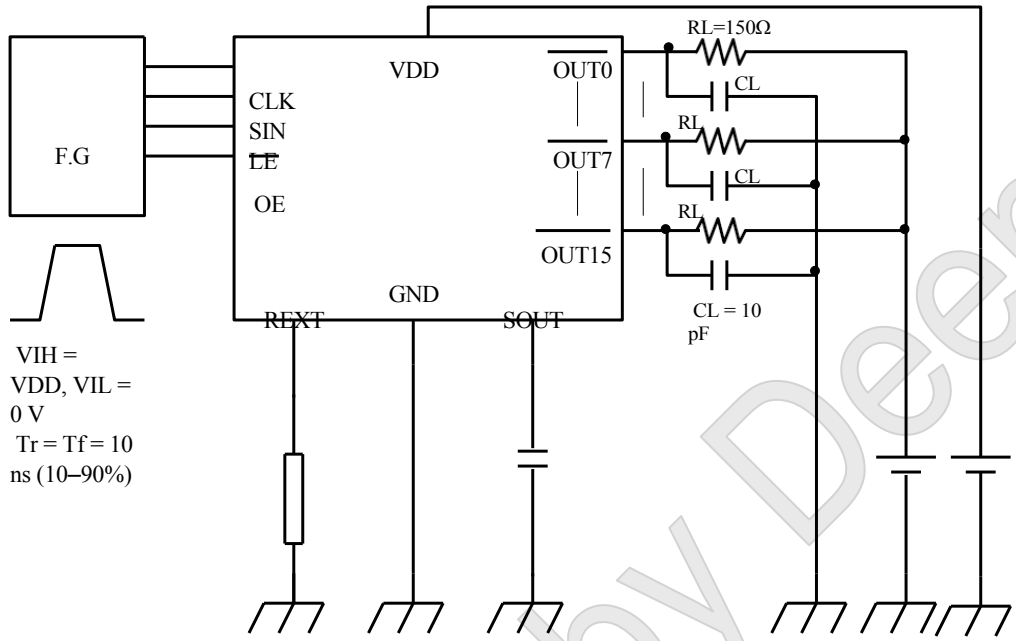
Constant current output Voltage Regulation	%VOUT	5	VDD=5.0V VO = 1.0–3.0 V, REXT = 1.18 kΩ	-	±0.1		%/V
Pull-up resistor	RUP	3	$\overline{OE}$	200	240	350	KΩ
Pull-down resistor	RDOWN	2	LE	250	340	450	KΩ

Switch Characteristics

Characteristics		Symbol	Test Circuit	Test Conditions	Minimum Value	Typical Value	Maximum	Unit
Transmission Latency	$\overline{OE}$—$\overline{OUT0}$	TPLH3	6	LE=H	-	25	40	nS
	$\overline{OE}$—$\overline{OUT1}$	TPHL3	6	LE=H	-	30	50	
	CLK-SOUT	TPHL	6	-	-	25	30	
Output Rise Time		TOR	6	10–90% of the voltage waveform	-	15	20	nS
Output fall time		TOR	6	90–10% of the voltage waveform	-	26	31	nS

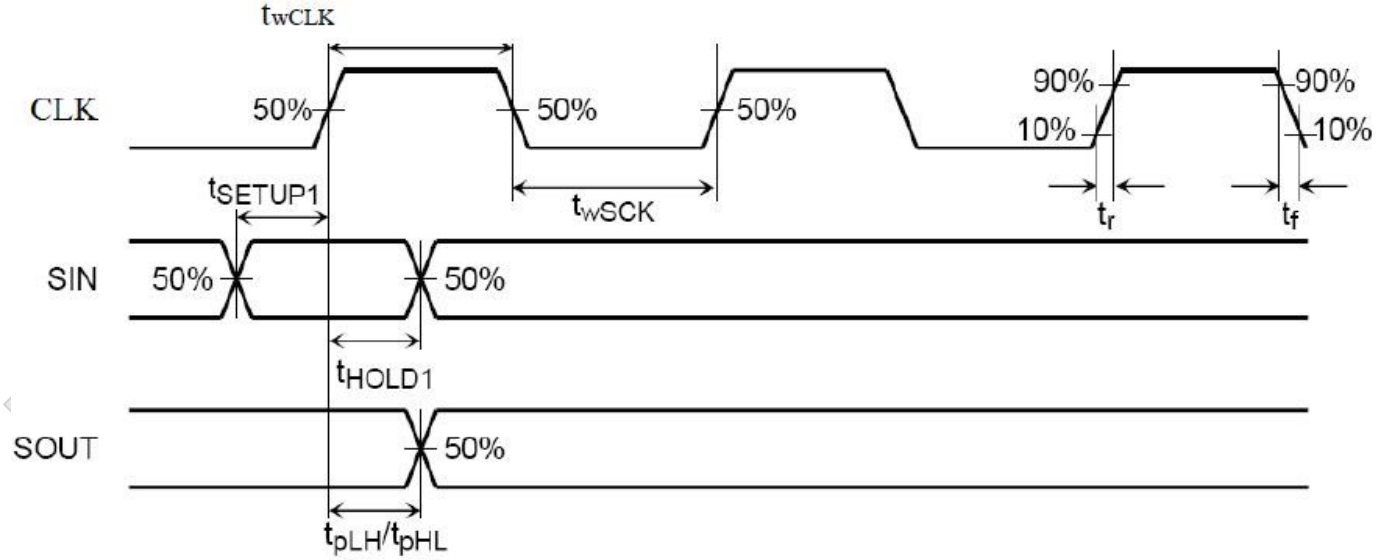
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Test circuit



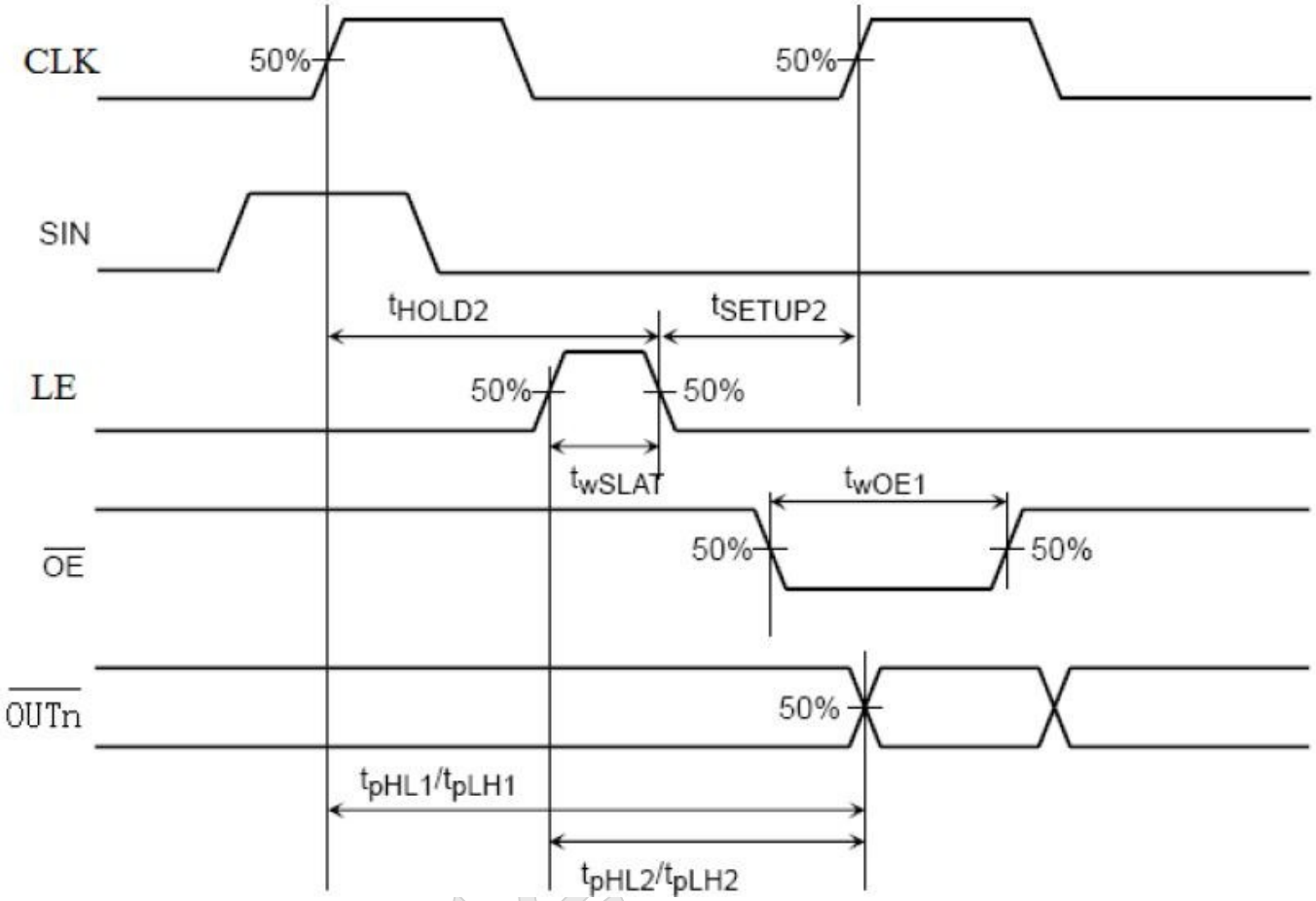
Timing Waveforms

1. CLK, SIN, SOUT

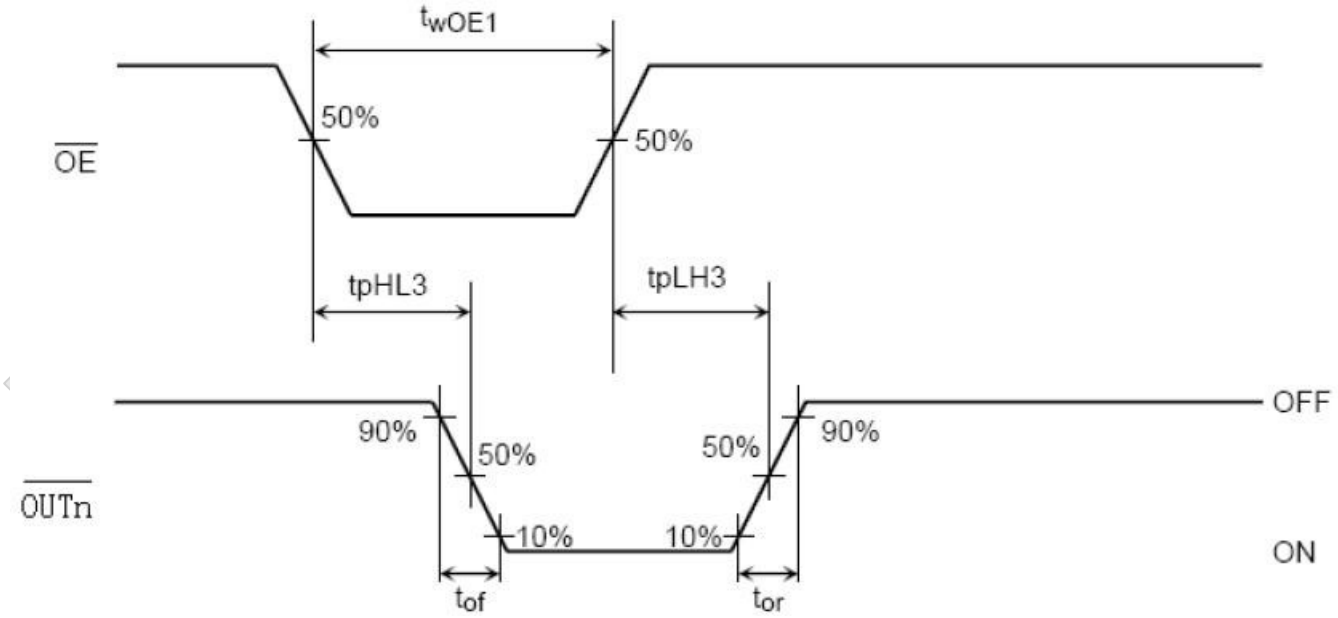


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2. CLK, SIN, LE, OE, OUTN



3. OUTN

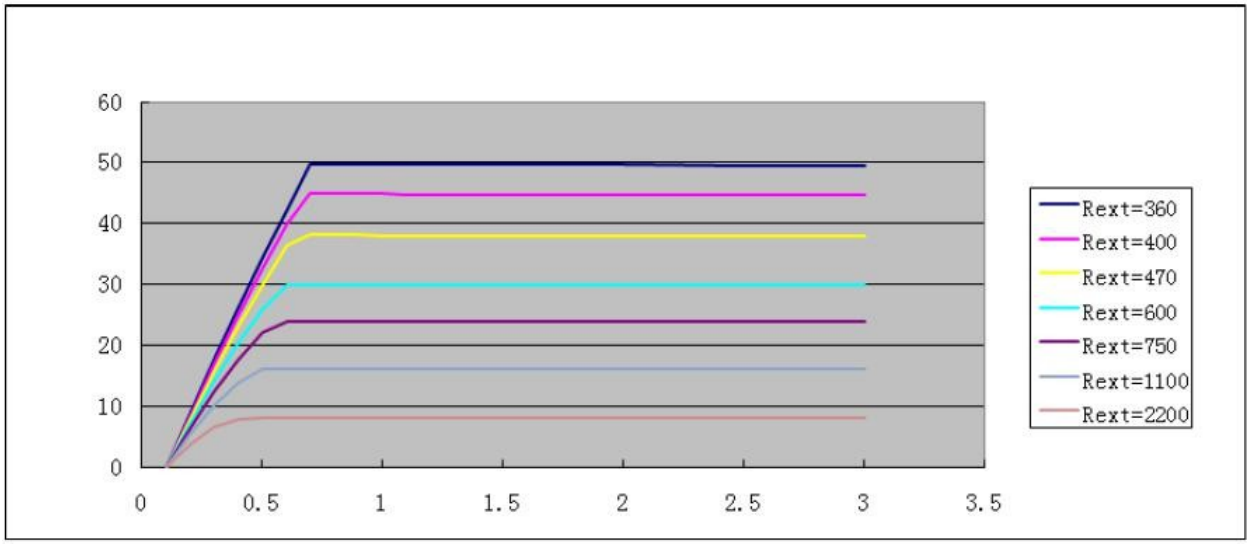


Application Information

The FM6124 employs precise current-drive control technology, resulting in minimal current variation between different channels on the same chip and between different chips.

1) The current variation between channels is <±2%, and the current variation between chips is <±3.5%.
2) It features current output characteristics that are unaffected by the load-side voltage, as shown in the figure below. The output current does not vary with the LED forward voltage V_F

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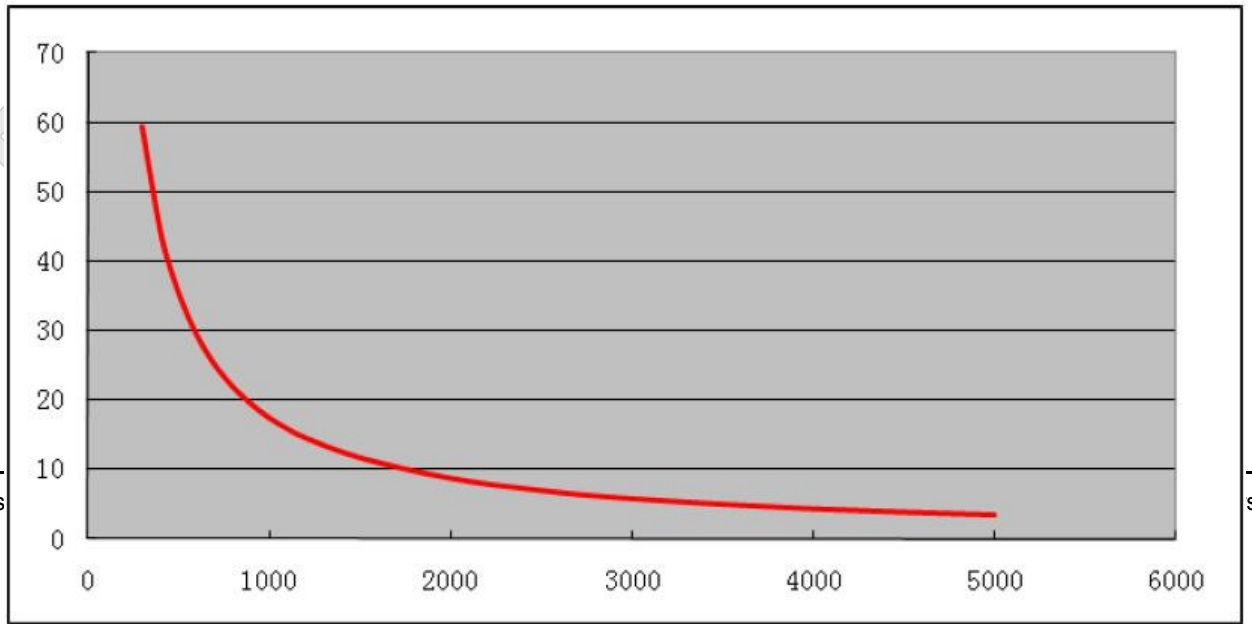


Adjusting the Output Current

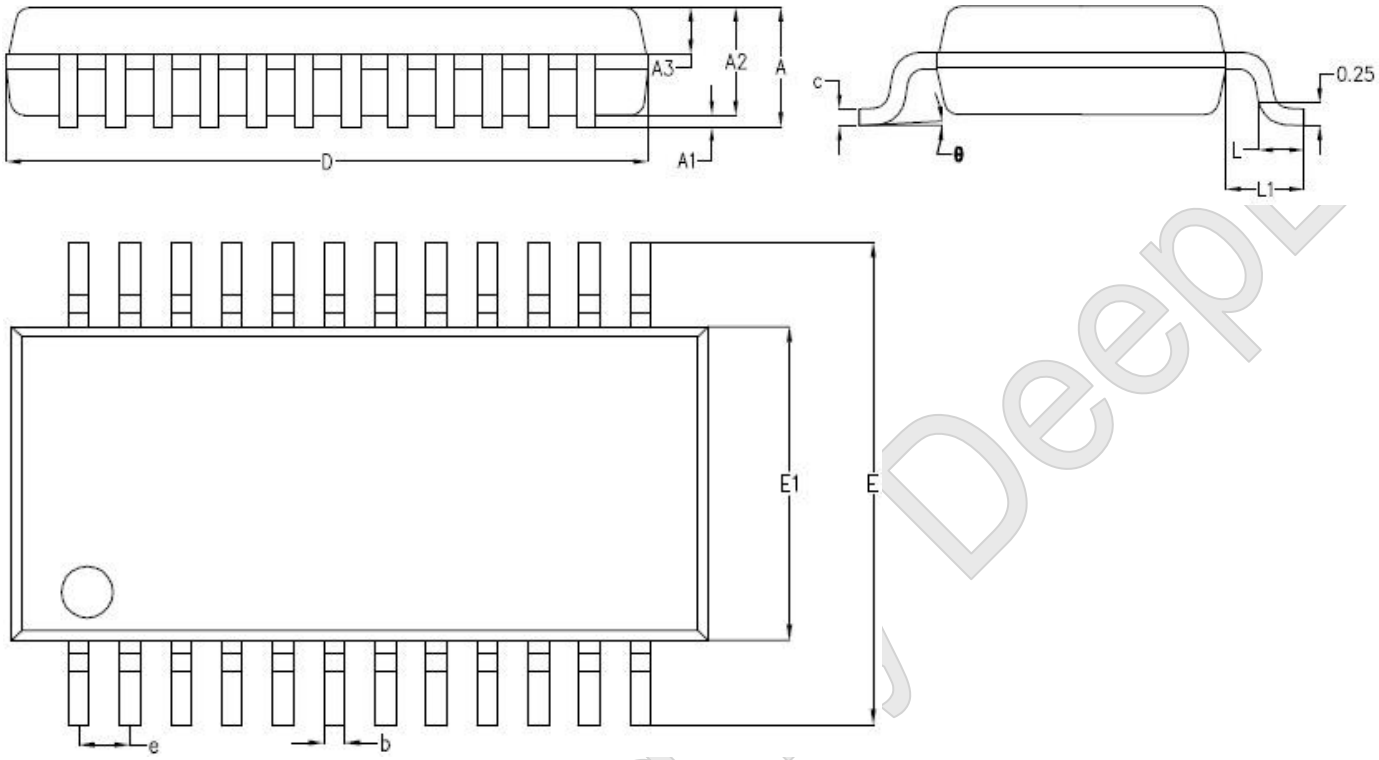
The FM6124 adjusts the output current (Iout) via an external resistor Rext, using the following formula:

VR-EXT = 1.191 V;

Iout = (V_{R-EXT} / Rext) * 15



Package Information



Symbol	millimeters		
	Minimum	Typical Value	Maximum
A	-	1.60	1.65
A1	-	0.15	0.20
A2	1.40	1.45	1.50
A3	0.60	0.65	0.70
b	0.22	0.25	0.30
c	0.17	0.22	0.25
D	8.55	8.65	8.75
E	5.90	6.00	6.10
E1	3.80	3.90	4.00
e	0.635 BSC		
L	0.57	0.60	0.65
L1	1.05 BSC		
θ	0°	3°	6°

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